

Cosmetic denialism and artificial intelligence: A new form of health misinformation

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To the editor,

Emerging communication patterns

In recent years, the spread of pseudoscientific content on social media has become increasingly structured and persuasive. Within aesthetic medicine and cosmeceuticals, a novel communication pattern is emerging that may be defined as “cosmetic denialism.” Unlike traditional misinformation, this phenomenon does not rely on entirely false claims but rather on the selective reinterpretation of real data. Documented adverse events and therapeutic limitations are removed from their clinical context and amplified, resulting in a distorted perception of risk. This dynamic reflects broader concerns related to the global “infodemic,” characterized by an overabundance of information that complicates public health decision-making¹.

The role of artificial intelligence

Artificial intelligence plays a pivotal role in this evolving landscape. It enables the generation of highly structured, visually refined, and apparently evidence-based content. However, such credibility is often superficial, arising from selective citation practices (cherry-picking) and the oversimplification of complex biomedical concepts². The communication pattern is consistent: a real data point is extracted, decontextualized, and amplified until it is perceived as systematic and inevitable.

Distortion of scientific doubt

A critical aspect of this phenomenon is the misuse of scientific doubt. Traditionally, doubt serves as a methodological tool to test hypotheses and strengthen



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evidence. In contrast, within these narratives, it is transformed into a rhetorical device aimed at generating selective mistrust and reinforcing pre-existing beliefs.

Clinical and communication implications

These dynamics are consistent with known mechanisms of misinformation amplification in digital environments, where simplified and emotionally charged content spreads more efficiently³. The implications are significant: increased patient confusion, erosion of trust in evidence-based medicine, and challenges in physician–patient communication. Importantly, the safety and efficacy of widely used aesthetic therapies, including botulinum toxin and hyaluronic acid fillers, are well established when performed appropriately by qualified professionals^{4,5}.

Conclusion

While artificial intelligence represents a powerful tool for scientific dissemination, it may also amplify misinformation when used without critical

competence. Promoting scientific literacy, contextual interpretation of data, and methodological rigor is essential to counteract these emerging pseudoscientific narratives.

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